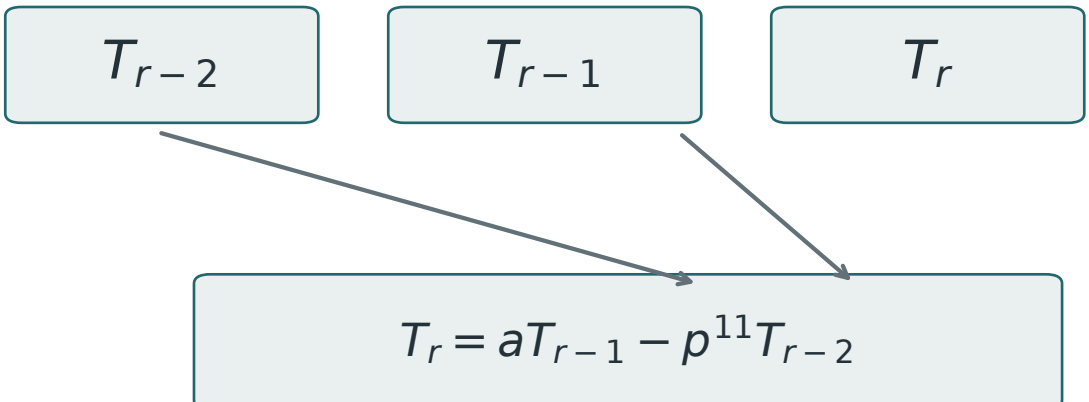

$$T_0 = 1, \quad T_1 = a = \tau(p)$$

Fixed prime p. Recurrence varies the exponent r, not the prime.